

Exercise 3 – Number System (Decimal to other number system)

Double Dabble

Convert a decimal number to any number system:

Step 1 : Divide the number by the base or radix of the target number system and note down **dividend** and **remainder**.

Step 2 : Repeat Step 1 until dividend is less than the base or radix

Step 3 : The resultant number is the digits represented by all the remainders in reverse order (see example below)

Example 1: Convert from **Decimal to binary (base 2)** number system

2		28	
2		14	0
2		7	1
2		3	1
		1	0

Answer = $(10110)_2$

Example 2: Convert from **Decimal to octal (base 8)** number system

8		128	
8		16	0
		2	0

Answer = $(200)_8$

Example 3: Convert from **Decimal to hexa decimal (base 16)** number system

16		268	
16		108	0
		6	12

Answer = $6(12)0 \Rightarrow (6C0)_{16}$

Questions:

Convert following decimal numbers to given number systems.

- (a) $201 \Rightarrow (?)_2$ (b) $965 \Rightarrow (?)_8$ (v) $123 \Rightarrow (?)_4$
(d) $1024 \Rightarrow (?)_{16}$ (e) $35 \Rightarrow (?)_8$ (f) $1103 \Rightarrow (?)_2$
(g) $888 \Rightarrow (?)_8$ (h) $321 \Rightarrow (?)_2$ (i) $4398 \Rightarrow (?)_{16}$